FF-D-396H
March 9, 1977
SUPERSEDING
Fed. Spec. FF-D-396G
February 7, 1972

FEDERAL SPECIFICATION

DISPENSERS, SOAP

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers rigid dispensers suitable for dispensing liquid, powder, and waterless paste soap.

1.2 Classification.

1.2.1 Types. Soap dispensers furnished under this specification shall be of the types indicated in table I.

1.2.2 Definitive part number (DPN). Soap dispensers covered by this specification shall be identified by a DPN. This part number is intended for cataloging and ordering purpose only. The DPN shall consist of the Federal Specification symbol plus the specification part number (SPN) (see table I). The DPN shall be written as indicated below (see 6.2).

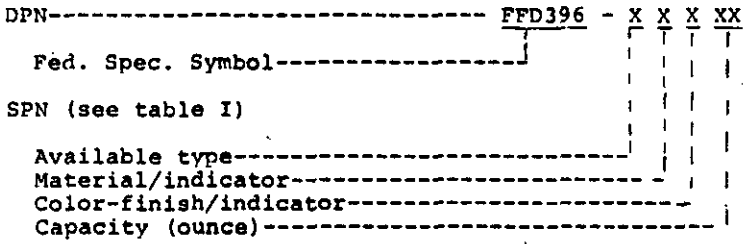


TABLE I. SPN identifier

SPN	Type 1/	Material/ indicator	Color-finish/ indicator	Capacity (ounce)
3SP20	III	Stainless steel/S	Polished/P	20
3CW20	III	Steel or plastic/C	Baked enamel-white/W	20
7XX10	VII	Any acceptable/X	Any acceptable/X	10
8XX40	VIIIB	do	do	40

1/ Type identification
III - Powder.
VII - Lather.
VIIIB- Paste, waterless (with integal refillable container).

FSC 4510

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

P-H-31 - Hand Cleaner, Duplicating Ink; Hand Cleaner, Grease.
P-S-577 - Soap, Grit, Hand, (Paste and Powder).
P-S-624 - Soap, Toilet, Liquid and Paste.
P-S-625 - Soap, Toilet, (Powdered: Plain and with Borax).
PPP-B-566 - Box, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Box, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza., Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Standard commercial product. Except as otherwise specified herein, soap dispensers covered by this specification shall be the manufacturer's standard commercial product for mounting on a flat surface.

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3.2 General.

3.2.1 Materials. Materials for dispensers shall be as specified hereinafter. All materials shall be of the quality and thickness used in normal practice for the intended purpose.

3.2.2 Construction.

3.2.2.1 Metal container. The metal container for liquid (lather) and paste soaps shall be of one piece and seamless, or shall be of two seamless sections securely joined together with the joints smoothly finished. The metal containers for powdered soap shall be formed or fabricated in accordance with the manufacturer's normal practice.

3.2.2.2 Soap dispensing components all dispensers. All moving valve or dispensing components that come in direct contact with the soap shall be of materials as follows:

a. Parts in contact with soap shall be of stainless steel or chromium plated carbon steel.

b. All springs shall be of stainless steel.

3.3 Cleaning and filling. Soap dispensers shall be capable of being cleaned and filled without the use of special tools. Dispensers shall not leak or drip when the container is filled to capacity.

3.4 Finish.

3.4.1 Exposed or visible metal parts. All exposed or visible metal parts except stainless steel and aluminum shall have a white baked enamel, white porcelain enamel or chromium plated finish as specified hereinafter. All exposed visible metal parts of stainless steel and aluminum shall have a general purpose polished finish, or better, conforming to the manufacturer's standard commercial product.

3.4.2 Exposed or visible plastics. All exposed plastic surfaces shall be chromium plated or white as specified by the SPN.

3.4.3 Internal metal parts. All internal metal parts, except stainless steel and aluminum, shall be either porcelain enamel, chromium plated, cadmium plated, or nickel plated.

3.5 Brackets and holders, types III and VII. A wall type bracket or holder and not less than three mounting screws shall be furnished with each type III and VII soap dispenser. Not less than three screw holes shall be provided in each bracket or holder. The brackets or holders shall be of substantial construction and designed to permit secure attachment to a flat, vertical surface, providing a flush, even contact between the wall and the dispenser. Brackets and holders shall be designed for either exposed or concealed installation. A mounting screw for each screw hole shall be furnished by the contractor.

3.6 Assembled unit. All dispenser components shall be furnished as completely assembled units, ready for mounting. Adjustable soap discharge mechanisms shall be factory adjusted to dispense the soap at the discharge rate hereinafter specified.

3.6.1 Replacing components. The moving parts for soap dispensers shall be easily removable and replaceable while the body of the dispenser remains mounted in its service position. Only common hand tools shall be required to effect the disassembly/replacement operation. This procedure, when conducted in accordance with manufacturer's instruction, shall not damage either the dispensing mechanism or the body of the dispenser.

3.7 Instruction. Instruction for installation, operation, and maintenance of the soap dispensers shall be provided with each dispenser. The instructions shall include repair procedure and a listing of available replacement parts.

3.8 Workmanship. Dispensers shall be fabricated and finished in accordance with the best commercial practices and shall be free of sharp edges and projections, burrs, ridges, seam lines, porosity, warpages, cracks, chipped edges, blisters, scratches, peeling and other defects affecting the appearance or which might affect the function or serviceability of the unit. Minor shrink marks not affecting function or appearance shall be permitted.

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3.9 In addition to the general requirements specified in 3.1 through 3.8, soap dispensers shall conform to the specific requirements of 3.10 through 3.12.6 as applicable.

3.10 Type III, powder.

3.10.1 Soap container material and finish. The soap container material shall be as specified for the SPN listed below. The material finish shall be in accordance with 3.4.

3.10.1.1 SPN 3SP20. The material shall be stainless steel.

3.10.1.2 SPN 3CW20. The material shall be carbon steel or plastic. The color of the plastic shall be white. The exterior finish for steel shall be white porcelain - or white baked on enamel.

3.10.2 Design. The type III soap dispenser shall be designed to dispense powdered soap conforming to P-S-577 and P-S-625. The dispenser shall be equipped with a soap feed mechanism and an agitator designed to dispense easily crushed caked and lumped powdered soap. The top cover or filling cap shall be secured by a tab-type latch, internal friction fit or other system that will retain the cap or cover in place during normal use.

3.10.2.1 Soap feed valve. The soap feed mechanism shall be either adjustable or non-adjustable at the manufacturer's option. The non-adjustable feed mechanism shall function in such a manner that approximately equal amounts of soap are dispersed with each operation. Adjustable soap feed mechanisms shall be readily adjustable to increase or decrease the amount of soap dispensed with each operation.

3.10.2.2 Agitator. Each operation shall actuate the agitator to prevent the build up of caked and lumped powdered soap.

3.10.2.3 Soap discharge opening. The soap discharge opening and dispensing mechanism shall be so arranged that the powdered soap will not be subjected to wetting during normal usage causing the soap to cake and restricting the operation of the mechanism.

3.10.2.4 Mounting brackets. See 3.5.

3.10.3 Capacity. The capacity of the soap container shall be not less than 20 fluid ounces or 38.0 cubic inches (see 4.5.1).

3.10.4 Bleeding. The dispenser shall show no signs of bleeding when tested in accordance with 4.5.3.

3.10.5 Discharge. The dispenser shall require not less than 400 nor more than 1200 positive actions to remove 12 fluid ounces (21.6 cubic inches) of powdered soap when tested in accordance with 4.5.5.1.

3.11 Type VII, lather; SPN 7XX10.

3.11.1 Material and finish. The soap container shall be formed of glass, metal, or plastic, or combinations thereof, at the manufacturer's option. Die castings of zinc alloy shall be acceptable as specified in 3.11.2. The material finish shall be in accordance with 3.4.

3.11.2 Design. The soap dispenser shall be designed to dispense liquid soap, conforming to P-S-624, in lather form, and shall have a "push-in" valve. The valve shall be operated with the aid of a spring arrangement. The top cover and valve body for the dispenser may be furnished as an integral unit or as separate components. Die castings of zinc based alloys shall be on acceptable material for the top cover and valve body.

3.11.3 Capacity. The capacity of the soap container shall be not less than 10 fluid ounces (see 4.5.1).

3.11.4 Leakage. No evidence of leakage shall be observed when the dispenser is tested in accordance with 4.5.2.

3.11.5 Bleeding. The dispenser shall show no signs of bleeding when tested in accordance with 4.5.3.

3.11.6 Discharge. The dispenser shall require not less than 800 nor more than 2000 positive actions to remove 10 fluid ounces of liquid soap when tested in accordance with 4.5.6.1.

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3.11.7 Lather valve test. The lather valve shall dispense 275 cc. of lather containing not more than 30 cc. nor less than 15 cc. of liquid soap when tested in accordance with 4.5.6.2.

3.11.8 Valve operating force. The force required for each positive actuation of the valve shall not exceed 8 pounds when tested in accordance with 4.5.6.3.

3.12 Type VIIIB, paste, waterless (with integral, refillable container) SPN 8XX40.

3.12.1 Material and finish. The soap container material shall be glass, metal, or plastic, or combinations of at the manufacturer's option. The material finish shall be in accordance with 3.4.

3.12.2 Design. The type VIIIB soap dispenser shall be designed to dispense waterless paste or cream hand cleaner conforming to P-H-31. The dispenser shall consist of a rigid dispenser body, incorporating an actuating device, a dispensing nozzle, and an integral refillable soap container. The dispensing mechanism and dispensing nozzle shall be designed for one hand operation. All moving components shall be simple, rugged, and lubricated by the cleaner itself.

3.12.2.1 Refillable soap container. The refillable soap container shall be designed for use with a follower plate or plunger which rests on top of the soap and follows the soap level down as the soap is dispensed. Provision shall be made to prevent air lock in the soap and feed system.

3.12.2.2 Mounting provisions. The dispenser shall be designed for secure attachment to a flat, vertical surface, providing a flush, even contact between the wall and the dispenser. Not less than three mounting holes and screws shall be provided.

3.12.3 Capacity. The capacity of the soap container shall be not less than 40 fluid ounces (see 4.5.1).

3.12.4 Leakage. No evidence of leakage shall be observed when the dispenser is tested in accordance with 4.5.2.

3.13.5 Bleeding. The dispenser shall show no signs of bleeding when tested in accordance with 4.5.3.

3.12.6 Discharge. The dispenser shall deliver an average of 2 grams $\pm$ 0.5 grams with each positive action of the dispensing mechanism when tested in accordance with 4.5.7.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Visual and dimensional examination of end item. Sampling for these characteristics shall be in accordance with inspection level S-3 in MIL-STD-105.

4.2.2 Samples for tests. Sampling for tests shall be in accordance with inspection level S-1 in MIL-STD-105.

4.3 Examination and tests.

4.3.1 Dimensional and visual examination. Soap dispensers selected in accordance with 4.2.1 shall be examined and measured for defects listed in table II. The Acceptable Quality Levels (AQL's) for defects shall be 2.5 percent defective for major defects and 6.5 percent defective for total defects.

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TABLE II. Classification of defects, end item

Defects	Major	Minor
SPN not as specified.	X	
Soap containers not formed of materials specified.	X	X
Soap container leaks when filled to capacity.	X	
External finish not as specified or required.	X	
Internal metal parts not finished as required.	X	
Mounting screws missing.		X
Components missing or units not fully assembled.	X	
Valves and moving parts not easily removable and replaceable.	X	
Instructions incomplete or missing.		X
Not free of warpage, cracks, or chipped edges.	X	
Finish blistered, scratched or peeling.	X	
Damage or defects affecting function or serviceability.	X	
Damage or defects not affecting function or serviceability.		XX

4.3.2 Test AQL. The AQL for tests shall be 4.0 percent defective.

4.4 Preparation of samples. Each soap dispenser selected in accordance with 4.2.2 shall be mounted to a rigid, vertical structure or wall in accordance with the manufacturers instructions.

4.5 Test methods.

4.5.1 Capacity. Fill the soap container with soap conforming to P-H-31, P-S-577, P-S-624, or P-S-625, as applicable. Observe for conformance to minimum capacity requirements (see 3.10.3, 3.11.3, and 3.12.3).

4.5.2 Leakage. Actuate dispensing mechanism until soap emerges from the dispensing opening. Place a catch basin beneath the unit. After 24 hours, examine dispenser and basin for evidence of leakage (see 3.11.4 and 3.12.4).

4.5.3 Bleeding. Actuate the dispensing mechanism one time (one cycle) as specified herein. At no point shall there be a free flow of soap from the soap dispensing opening. A definite amount of soap shall be discharged and the discharge thereafter shall cease. Five test cycles shall be conducted with a 30 second minimum between cycles (see 3.10.4, 3.11.5, and 3.12.5).

4.5.4 Discharge (general). Actuate the dispensing mechanism the travel distance at an approximately uniform rate. Record the number of actuations required to empty the container (see 3.10.5 and 3.11.6).

4.5.5 Type III, powder, SPNs 3SP20 and 3CW20.

4.5.5.1 Discharge. Fill dispenser with amount of soap specified in 3.10.5. The powdered soap shall conform to P-S-577 or P-S-625. Test in accordance with the procedure described in 4.5.4 (see 3.10.5).

4.5.6 Type VII, lather, SPN 7XX10.

4.5.6.1 Discharge. Fill dispenser with amount of soap specified in 3.11.6. The soap shall conform to P-S-624. Test in accordance with the procedure described in 4.5.4 (see 3.11.6).

4.5.6.2 Lather valve test (see 3.11.7).

a. Preparation of sample. Immediately after performing the discharge test in 4.5.6.1, refill the soap container with liquid soap conforming to P-S-624.

b. Procedure. Take a clean, dry, standard 250 cc. laboratory beaker (275 cc. level full). At the rate of approximately 80 strokes per minute, pump lather into the beaker, working the beaker around so that it is filled with lather and there are no large air spaces. Allow beaker to stand until all lather has condensed into liquid. This may be hastened by adding alcohol slowly, drop by drop, to the lather from a pipette. Measure the alcohol (it will take about 3cc.) to entirely disintegrate the lather. After the lather has turned completely to liquid, measure the amount of liquid. If alcohol is used, deduct the amount of alcohol that was added. Compute the net amount of liquid soap for conformance to 3.11.7.

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4.5.6.3 Valve operating force. Empty the soap from the container and remove the dispenser from its mounting. Securely mount the dispenser with the valve vertical. The force required to operate the valve shall not exceed that specified in 3.11.8.

4.5.7 Type VIIIB, SPN 8XX40.

4.5.7.1 Discharge (see 3.12.6).

a. Preparation of sample. Prepare the dispenser for operation in accordance with the manufacturer's instructions. Fill or load the dispenser, as applicable, with waterless hand cleaner conforming to P-H-31. Actuate the dispensing mechanism at a uniform rate until all of the priming oil or grease that may coat mechanism parts is removed and dispenser is fully primed.

b. Procedure. Take a clean, dry, petri dish, and weigh on a gram scale. Dispense soap into the petri dish with ten positive actions of the dispensing mechanism and reweigh dish and soap. Subtract the weight of the petri dish to find the net weight of the soap. Determine the average weight of soap delivered per pumping action by dividing the net weight of the soap by ten.

4.6 Inspection of preparation for delivery. An inspection shall be made to determine whether the packaging, packing and marking requirements comply with section 5 of this specification. Defects shall be scored in accordance with table III. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE III. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each dispenser shall be packaged in a close-fitting box conforming to PPP-B-566 or to PPP-B-636. Closure of boxes shall be in accordance with the appendix to the applicable box specification.

5.1.2 Commercial. The soap dispensers shall be packaged in unit containers normally used in commercial practice.

5.2 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Dispensers packaged as specified in 5.1 shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type. The contents shall be blocked and braced to prevent movement inside the box. Closure and strapping of the boxes shall be in accordance with the box specification.

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5.2.2 Commercial. The packaged soap dispensers shall be packed in shipping containers to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Unitization. (Applicable to full rail car and truck load shipments only.) The packed soap dispensers shall be unitized for shipment in accordance with normal commercial practice. The unitized load shall not exceed 2,500 lbs. in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.4 Marking. Packages, shipping containers, and unitized loads (when applicable) shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. Soap dispensers covered by this specification are intended for use as dispensers of liquid and powdered soap and paste type waterless hand cleaner.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Definitive part number (see 1.2.2).
(c) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

MILITARY INTEREST:

Custodians:

Navy - YD
Air Force - 84

Review activity:

DCSC - CS

User activity:

Navy - MC

Preparing Activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

DC GOVT-DCG
HEW-FEC
GSA-FSS

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